MSNet: Self-Supervised Multiscale Network With Enhanced Separation Training for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (HAD) has attracted increasing attention due to its economical and efficient applications. The main challenge lies in the data-starved problem of hyperspectral images (HSIs) and the costliness of manual annotation, making it heavily reliant on the model's adaptability and robustness to unseen scenes under limited samples. Self-supervised learning offers a solution to this urgency via mining meaningful representations from the data itself. One promising paradigm is leveraging untrained neural networks to reconstruct the background component for revealing anomalous information. Its capability stems from the network architecture and the training process rather than learning from expensive and strongly domain-dependent data, which is naturally applicable to HAD. In this article, to handle the urgent requirement for self-supervised learning in HAD, we propose a multiscale network (termed MSNet) that detects anomalies with enhanced separation training. The network architecture consists of several multiscale convolutional encoder–decoder (CED) layers, considering the spatial characteristics of the anomalies. To suppress the anomalies during background reconstruction, we adopt a new separation training strategy by introducing a soft separator for better practicality on larger datasets. Extensive experiments conducted on five commonly used datasets and the HAD100 dataset, demonstrate the superiority of our method over its counterparts. Our code is available at https://github.com/enter-i-username/MSNet.**

***Index Terms*— Enhanced separation training, hyperspectral anomaly detection (HAD), multiscale network, self-supervised learning.**

NOMENCLATURE

$\mathbf{X}$	Network input hyperspectral image.
$\hat{\mathbf{X}}$	Network output background image.
$\mathbf{M}$	Binary mask. Ones indicate the locations of anomalous pixels, while zeros represent the background pixels.
$\bar{\mathbf{M}}$	Boolean *not* operation on $\mathbf{M}$.
ϵ	Small constant that prevents division by zero.
$S(\cdot)$	Summation of all elements in the input matrix.
$\odot$	Element-wise multiplication over spatial dimensions.
$\|\cdot\|_F$	Frobenius norm.

Manuscript received 17 May 2024; accepted 4 June 2024. Date of publication 7 June 2024; date of current version 20 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U2133211, Grant 62001063, and Grant 61971072; and in part by the Fundamental Research Funds for the Central Universities under Grant 2023CDJXY-037. *(Corresponding author: Xichuan Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2024.3411009

I. INTRODUCTION

HYPERSPECTRAL imagery (HSI) has been receiving great attention in the field of remote sensing over the past decades and has been widely utilized in various tasks [1], [2], [3], [4], [5], [6]. Among these studies, hyperspectral anomaly detection (HAD) stands out as a crucial technique for localizing targets of interest in HSIs without the need for human supervision [7]. This is particularly relevant due to the data-starved problem of HSIs [8], [9] and the costliness of obtaining human labels. Therefore, HAD can be applied economically and efficiently for various civilian and military purposes [10], [11], [12]. Hyperspectral anomalies are defined as an extremely small number of pixels with spectral signatures that differ significantly from the surrounding or global background [13]. Generally, the imbalance between the samples of background and anomalies makes the direct binary classification of abnormal pixels against background pixels quite challenging [14]. More importantly, due to the diversity of spectral sensors and anomalous targets, achieving consistent performance across different scenarios with a unified learning model is difficult, placing high demands on the model's adaptability and robustness. To date, various studies have proposed solutions to address this difficult yet crucial problem from different perspectives. These solutions can be broadly categorized into four types: unsupervised background modeling, semi-supervised background reconstruction, weakly supervised discriminative learning, and self-supervised background reconstruction.[1]

Unsupervised methods model the background from the observed HSI in a model-driven manner, leveraging prior knowledge and assumptions from the background component. They measure the abnormality of each pixel by calculating the difference between the original image and the background. In 1990s, Reed and Xiaoli proposed the RXD [15], which

[1]The methods of four types all possess an unsupervised nature since they do not require human-annotated labels. In this article, we categorize them based on their respective methodologies. For example, the self-supervised methods take the image itself as the supervision information, and the semi-supervised methods only screen background pixels as training samples.

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assumes that the background pixels follow a multivariate Gaussian distribution and utilizes the Mahalanobis distance as a measure of the deviation of outliers from the background. Building upon this foundation, various enhancements have been developed, such as the local information-based RXD [16], the kernel trick-driven RXD [17], and other representatives [18], [19], [20]. In addition to these probability distribution-based methods, some others model the background using man-made image priors, also referred to as representation-based methods [13]. They hold a fundamental assumption that background pixels can be represented by background pixels locally or globally, whereas anomalous pixels cannot [21]. For example, the collaborative representation-based detector (CRD) [22] considers each background pixel as a linear combination of its neighboring pixels, which are included in a local dual window. To exploit more characteristics of the HSI from a global view, two image priors, i.e., the low-rank property and the sparsity, are discussed in many studies [23], [24], [25], [26]. While these unsupervised model-driven methods maintain ideal and simplistic assumptions for background modeling, they may fail to accurately detect anomalies concealed within more complex scenes [27].

The remaining three types of methods conduct detection within a data-driven paradigm based on parameterized neural networks [28], [29], [30] rather than strongly relying on handcrafted priors, which are more practical and adaptive for handling different complex backgrounds. They train networks to learn low-level and abstract representations of the image and detect anomalies through reconstruction errors. The semi-supervised methods [31], [32] search for pure background samples in the image before training using clustering [33], [34] or distance measurement [35]. Different from semi-supervised ones, the weakly supervised group [27] automatically generates supervision information for both anomaly and background, holding a belief that the prior search is not accurate. It employs the generative adversarial network (GAN) [36] for discriminative learning between the coarsely annotated background and anomalous pixels. To conclude, the goal of the above-mentioned two types is to "overfit" their networks to a highly compact background boundary, exposing outliers far when testing [37]. Unfortunately, the boundary is not robustly guaranteed, especially in cases where the initial coarse searching mistakenly includes anomalies in the background set or vice versa.

Self-supervised methods treat the entire image as supervision information and aim to reconstruct the image itself using an untrained network beforehand, a similar way as in those image restoration tasks [9], [38], [39]. It operates under the assumption that learning the background is a more straightforward shortcut for the networks rather than outliers. Typically, the training samples are all pixels (spectral vectors) in the test HSI for autoencoders (AEs) [28], [40], [41] and patches for convolutional neural networks (CNNs) [42], [43]. Recently, the AETNet [44] introduced an image-level paradigm that takes normal HSIs as external training samples, and trained a unified network with anomaly data augmentation. This method has an incredibly high detection speed because it only needs inference when testing. However, due to limited diversity of the training set, this detector may be trapped within the distribution [45]. To make models more adaptive and robust to unseen scenes when data are limited, a remedy is to use an untrained network in a self-supervised manner [45]. Specifically, its capability stems from the network architecture and the training process (or objectives), but not from the network parameters that are randomly initialized before each test image instead of learning from external domain-dependent data. Both of the factors are prior to unseen data and determine how to mine meaningful representations from the data itself. Therefore, careful designs for the *network architecture* and the *training process* are of vital importance.

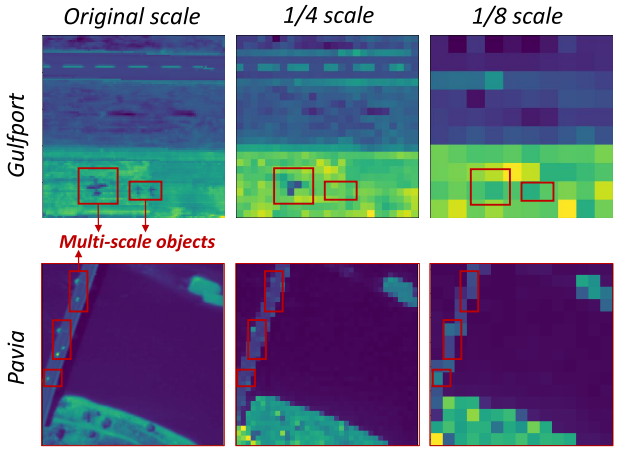


Fig. 1. Feature maps of three downsampling scales on the Gulfport and Pavia datasets. The Gulfport and Pavia datasets shown in the left column, respectively, contain airplanes and vehicles as anomalies of interest (boxed in red) in the imaging scenes. Due to the differences in their spatial resolutions (for different images) and object sizes (for the same or different images), these anomalies exhibit multiscale characteristics, posing a challenge to a unified single-scale learning model for background reconstruction. The downsampling operation is used to eliminate the anomalies in the background. A single finer downsampling can eliminate most vehicles in the Pavia dataset shown in the middle column, but it cannot handle the largest airplane in the Gulfport dataset. A single coarser downsampling can eliminate most anomalies as shown in the right column, but it also removes the background details. To reconstruct a better background for anomaly detection, the most robust way is to efficiently integrate all downsampled multiscale features to overcome these flaws.

For the network architecture, the key is to find an optimal structure that "resonates" with the inherent structure of data [38], while also maintaining low computational complexity. Currently, most AE-based methods [28], [41] mainly concentrate on the spectral features of the scene, which do not make full use of the spatial information. To handle the spatial structure of anomalous targets, Auto-AD [42] assumes the anomalies are extremely small in size and thus can be eliminated through the downscaling encoding during background reconstruction. However, anomalies of interest like airplanes or vehicles, as shown in Fig. 1, often exhibit multiscale characteristics. This problem arises due to two primary factors: first, anomalies may exhibit diverse shapes; second, imaging sensors may possess varying physical resolutions, resulting in inconsistencies in the scale of anomalies across different images. These factors pose a significant challenge to the model's ability to perform effectively across multiple scales. This inspired us to consider a multiscale downsampling architecture, which

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exploits the topological structures of different anomalies via multiple scale branches.

For the training process, one main concern is that background reconstruction is a unguided process due to the ambiguity in determining when to stop the training [41]. However, the detection performance via background reconstruction is sensitive to the timing of termination, rendering it unsuitable for automated detection on test images. In this context, directly taking the original image as the reconstructive target leads to an identical mapping problem (IMP) [46], because the network has a risk of learning a trivial identity mapping when converged. The potential solution documented in [46] is that the network first estimates the threshold from the background and anomaly components and subsequently performs separation training until the network converges to a dynamic equilibrium around the threshold. While the separation training strategy can effectively suppress anomalies and avoid the IMP, its effectiveness relies heavily on the availability of a precise threshold value, which is hard to estimate and requires fine-tuning for each specific image, and thus restricts its automated detection.

In this article, we present MSNet, a multiscale network featuring enhanced separation training for self-supervised HAD, to address the problem of multiscale anomalous targets and the IMP without threshold fine-tuning. MSNet is composed of several stacked convolutional encoder–decoder (CED) layers, each of which serves as a different spatial downscaling feature learner and eliminates the anomalies at its respective scale. The encoded and decoded features are fused with skip connections. Distinct from the classic multiscale network U-Net [47], we do not fuse the multiscale features using channel-wise concatenation, but use a summation for the fusion purpose. In addition, we keep the channel number of these features equal to the input, and apply a fast and efficient band selection algorithm OPBS [48] to the input HSI to further reduce total convolution kernels while preserving the informativeness and distinctiveness of the HSI. To further preserve the multiscale characteristics in both shallow and deep layers, similar to certain deblurring approaches [49], [50], we utilize a multiscale reconstruction loss as the content loss rather than the mean squared error (mse) loss.

Relying solely on the multiscale network cannot well address the widespread IMP in self-supervised HAD [28], [41], [42] because of the contamination of anomalous targets during background reconstruction. Therefore, we introduce the separation training to address this problem and further enhance it for flexible and robust results. The enhanced separation training includes two components: *the enhanced separation training loss* and *the latent mask updating*. The enhanced separation training loss relies on a latent mask to partition the HSI into background and anomaly parts, suppressing the anomaly while reconstructing the background to obtain an estimate of pure background. The latent mask is periodically updated during training by soft thresholding to adaptively correct the separation results and prevent contamination by anomalous targets. The softened updates remove the dependence on threshold parameters, resulting in more robust performance on larger datasets.

Our contributions can be summarized as the following.

1) We propose a multiscale network architecture, designated as MSNet, for the purpose of addressing the intricate issue of multiscale anomaly shapes. The MSNet framework incorporates multiple CED layers, each featuring distinct spatial downscaling factors, aimed at minimizing the influence of varying anomaly sizes. To facilitate efficient feature integration, we employ skip connections, which contributes significantly to the generation of accurate background images. Furthermore, to preserve the multiscale features throughout all layers, we incorporate a multiscale reconstruction loss as the content loss, ensuring that information from various scales is effectively utilized and propagated within the network.
2) We develop an enhanced separation training algorithm that aims to address the IMP within the self-supervised learning framework. This algorithm offers greater flexibility and eliminates the need for threshold fine-tuning. To mitigate the potential anomalies during background reconstruction, we incorporate an enhanced separation training loss. Additionally, to minimize the influence of the threshold parameter on the training process, we construct the update rule of the latent mask by introducing the soft mask paradigm, enabling a more gradual and adaptive adjustment of the mask.

We organize the rest of this article as follows: Section II reviews the IMP and a potential solution, separation training, both highly relevant to our method. Section III presents our method in detail, including the MSNet structure and the enhanced separation training. Extensively conducted experiments are described in Section IV. We conclude this article in Section V.

II. Preliminaries

Self-supervised methods regardless of their network structures, all suffer from unguided training due to their reconstruction nature, which may cause a severe performance decline in detection. We first review this problem and introduce a potential solution to it.

A. Identical Mapping Problem

Self-supervised detectors [28], [41], [42], [46] take the entire HSI as the supervision information, considering the background as valuable and easily learnable information for the network. By leveraging this understanding, these detectors efficiently filter out abnormality, focusing on learning and identifying the underlying patterns in the background. However, such detectors often assimilate both background and anomalies into the estimated background as the training progresses, behaving like identity mapping functions. This phenomenon is referred to as the IMP [46], resulting in a significant decline in detection performance. It is mainly caused by the unguided training process that the complex background is hardly reconstructed because of anomaly component included in the process.

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Fortunately, this issue can be addressed by using different reconstruction objectives. For example, the sparse AE method [28] constrained the sparsity of hidden codes, mitigating the IMP by limiting the model's capacity. RGAE [41] assumed the sparsity in anomalous targets, resulting in a slower convergence speed under the reconstruction with the $l_{2,1}$-norm compared to the background. Similarly, AutoAD [42] introduced an adaptive-weighted loss to regulate the convergence speed of pixels with larger reconstruction errors. Differently, BiGSeT [46] shifted the objective to reconstructing the background and completing potential anomalies with a process termed *separation training*. Rather than approximating the input HSI, the separation training considers the convergence as a dynamic equilibrium around a preestimated threshold between the background and anomalies. However, this training strategy can efficiently prevent reconstructing the anomalies only when a precise threshold is obtained, thereby limiting its flexibility.

B. Separation Training

Different from the unguided training process, the separation training algorithm introduces a latent binary mask explicitly dividing the pixels in the reconstructed image into two mutually exclusive parts, i.e., the potential anomaly part and the background part. By separately suppressing (i.e., removing and completing) the anomaly part while reconstructing the background part during training, the networks avoid anomalous component in the reconstructed image, and thus maintain high performance throughout the training.

Concretely, we train the network using the separation training loss $\mathcal{L}_{\text{SeT}}$ given below

$$\mathcal{L}_{\text{BR}} = \frac{1}{S(\bar{\mathbf{M}})}\|(\hat{\mathbf{X}} - \mathbf{X}) \odot \bar{\mathbf{M}}\|_F^2 \tag{1}$$

$$\mathcal{L}_{\text{AS}} = \frac{1}{S(\mathbf{M}) + \epsilon}\|\text{LoG}(\hat{\mathbf{X}}) \odot \mathbf{M}\|_F^2 \tag{2}$$

$$\mathcal{L}_{\text{SeT}} = \mathcal{L}_{\text{BR}} + \lambda \cdot \mathcal{L}_{\text{AS}} \tag{3}$$

where $\mathcal{L}_{\text{BR}}$ and $\mathcal{L}_{\text{AS}}$, respectively, indicate the background reconstruction loss and anomaly suppression loss, and λ is a tradeoff parameter in (3); $\text{LoG}(\hat{\mathbf{X}})$ in (2) represents the second-order Laplacian of Gaussian (LoG) operator applied to the estimated image $\hat{\mathbf{X}}$, which is computed as follows:

$$\mathbf{L} = \begin{bmatrix} -2 & -4 & -4 & -4 & -2 \\ -4 & 0 & 8 & 0 & -4 \\ -4 & 8 & 24 & 8 & -4 \\ -4 & 0 & 8 & 0 & -4 \\ -2 & -4 & -4 & -4 & -2 \end{bmatrix} \tag{4}$$

$$\text{LoG}(\hat{\mathbf{X}}) = \mathbf{L} * \hat{\mathbf{X}} \tag{5}$$

where $\mathbf{L}$ is a 5×5 LoG template, and $*$ computes 2-D convolutions over each band. The remaining notations in (1) and (2) are comprehensively elucidated in Nomenclature. It can be known from $\mathcal{L}_{\text{BR}}$ and $\mathcal{L}_{\text{AS}}$ that we reconstruct the background pixels selected by $\bar{\mathbf{M}}$, and we suppress the anomalous pixels selected by $\mathbf{M}$ by constraining the second-order spatial variation.

Algorithm 1 Separation Training Algorithm

Input: HSI data $\mathbf{X}$, tradeoff hyperparameter $\lambda > 0$, proportion threshold τ for binarization, number of training iterations $K \geq 1$, number of epochs in one iteration $E \geq 1$.

Initialize: $\mathbf{M}^{(0)} = \mathbf{0}$.

```
for m = 1 : K do
    for n = 1 : E do
        Feed the network with X ⊙ M̄^(m−1) as input and yield X̂^(m);
        Calculate L_SeT using (3);
        Do backward propagation;
    end for
    Calculate R^(m)_{i,j} = ‖X̂^(m)_{i,j,:} − X_{i,j,:}‖_2^2;
    Sort the elements in R^(m) in ascending order to create r̃^(m), and obtain the τ quantile r̃^(m)_(τ) from it;
    Calculate M^(m) using (6);
end for
```

Output: Detection map $\mathbf{R}^{(K)}$.

The latent variable $\mathbf{M}$ is predicted and periodically updated in our training process. Specifically, we divide the whole training into K iterations. In the mth iteration, we first train the network for a small number of epochs E using $\mathcal{L}_{\text{SeT}}$ with $\mathbf{M}^{(m-1)}$ derived from the previous iteration. Then, the mask is updated to $\mathbf{M}^{(m)}$ by the following binarization:

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{R}_{i,j} > \tilde{r}_{(\tau)} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where $\mathbf{R}$ denotes the reconstruction error matrix in this iteration, obtained by $\mathbf{R}_{i,j} = \|\hat{\mathbf{X}}_{i,j,:} - \mathbf{X}_{i,j,:}\|_2^2$; vector $\tilde{r}$ contains values sorted in ascending order from the matrix $\mathbf{R}$ and $\tilde{r}_{(\tau)}$ represents the τ quantile with a predefined proportion threshold $\tau \in (0, 1]$. The threshold parameter τ here means the proportion of anomalies over the total number of pixels and represents ones in $\mathbf{M}$. We set $\mathbf{M}^{(0)} = \mathbf{0}$ as an initial step, and the training algorithm is given in Algorithm 1.

In conclusion, pixels below τ are well reconstructed and those above τ will stay continuously suppressed, and thereby exists an equilibrium around τ. Although the termination condition is thus relaxed by fixing a proper K, the threshold has an unignorable impact on the performance and often needs fine-tuning.

III. Proposed MSNet

A. Overview

Our proposed method works under a self-supervised paradigm, whose detection performance is highly relevant to the network structure and the training process. Considering the multiscale characteristics of the anomalies across different images, we build our network with multiple downscaling branches to adapt to various possible scenes. In addition, we enhance the separation training strategy as stated in Section II to improve its robustness to larger datasets, by expanding the Boolean values of the latent mask to real values and removing the threshold parameter. Fig. 2 illustrates the overall structure of our proposed method.

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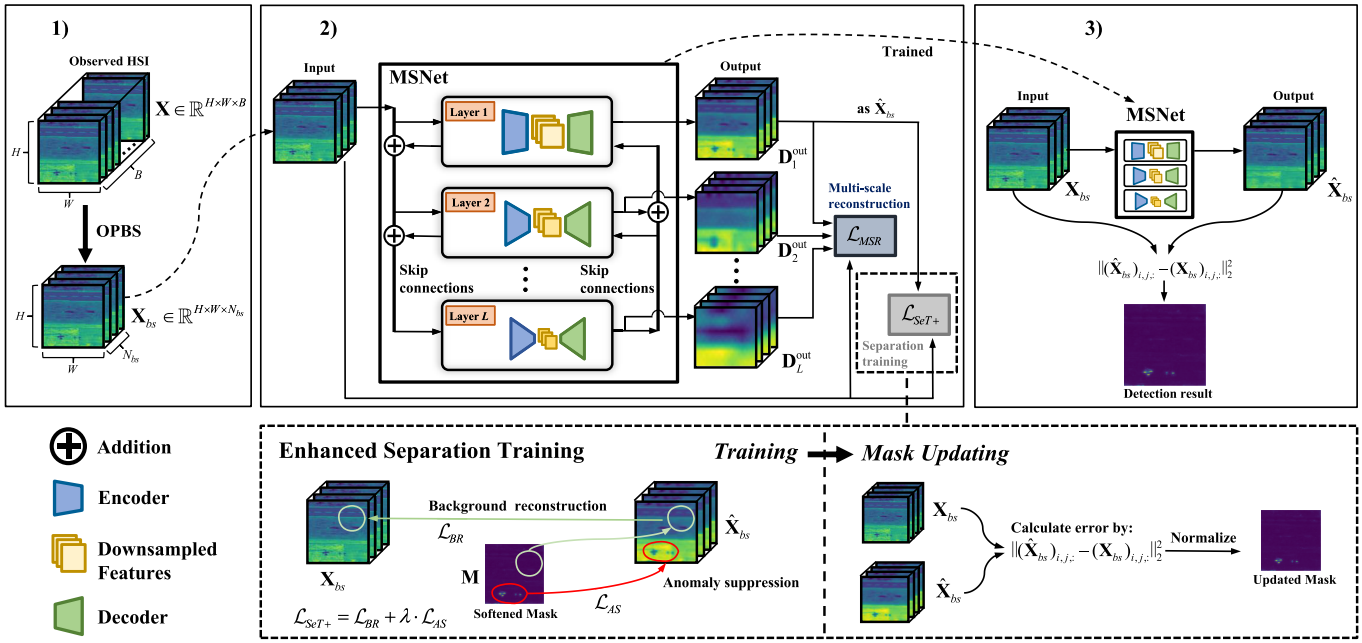


Fig. 2. Overview of our proposed MSNet method. Our detection consists of three steps. 1) Band selection, 2) multiscale network training, and 3) network inference to obtain the detection result. We use OPBS before training to remove band redundancy from the input HSI while preserving its discriminability. The MSNet architecture consists of L stacked multiscale layers, each of which serves as a downscaling feature learner at a different scale. Skip connections in summation trunks are employed to fuse both encoding and decoding features. We train the network using two losses including the multiscale reconstruction loss $\mathcal{L}_{\text{MSR}}$ and the enhanced separation training loss $\mathcal{L}_{\text{SeT+}}$. The output decoded features $\mathbf{D}^{\text{out}}$ are regarded as different scale levels of $\mathbf{X}_{\text{bs}}$ constrained by $\mathcal{L}_{\text{MSR}}$ as the content to retain both deep and shallow multiscale characteristics in the network. The topmost feature $\mathbf{D}_1^{\text{out}}$ is designated as the reconstructed background and undergoes separation training with $\mathbf{X}_{\text{bs}}$. In enhanced separation training, we use an adaptively updated soft latent mask to separately represent background and anomaly regions, reconstructing the background part while suppressing the anomaly part without the need for threshold fine-tuning. The final detection result is obtained by computing the error between the input image and the background computed from network inference.

Our network consists of several stacked CED layers, each serving as a background learner at a different scale. Under this setting, anomalies of different sizes can be eliminated during background reconstruction, rather than a single-scale architecture that might not be robust to various cases. Different from the U-Net, we fuse multiscale features extracted by CED layers through skip connections simply using summation, instead of channel-wise concatenation that might lead to catastrophic complexity. To further reduce the computational cost, we apply a fast and efficient band selection algorithm OPBS [48] as a pretraining step. This algorithm quickly selects an informative and distinctive subset of the original bands. Therefore, the complexity of the network can be reduced while the detection performance is maintained.

To train our network, two loss functions are used. The first multiscale reconstruction loss is to retain multiscale features within the deep layers as our content loss. The second enhanced separation training loss is to mitigate the IMP, preserving the background and suppressing the anomalies, completely independent of any preestimated threshold parameter.

Our detecting procedure consists of the following three steps: 1) band selection; 2) network training; and 3) network inference. Formally, let the observed HSI be denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent height, width, and spectral bands, respectively. After selecting a subset of N_{bs} bands out of B, we denote the subimage as $\mathbf{X}_{\text{bs}} \in \mathbb{R}^{H\times W\times N_{\text{bs}}}$. This subimage is subsequently fed into our network for training. In the training phase, we use the multiscale reconstruction loss and the enhanced separation training loss, following the separation training strategy. Finally, the well-trained network is used to estimate a pure background $\hat{\mathbf{X}}_{\text{bs}}$ by inference, and the detection map is computed via reconstruction errors between $\hat{\mathbf{X}}_{\text{bs}}$ and $\mathbf{X}_{\text{bs}}$.

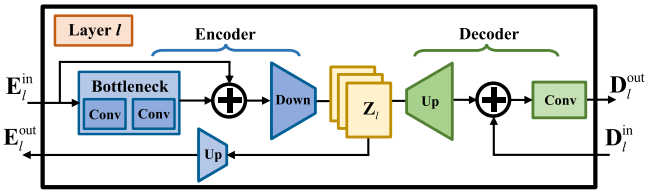


Fig. 3. lth CED layer. The input feature $\mathbf{E}_l^{\text{in}}$ of layer l first undergoes a bottleneck module and residual skip processing, followed by a down block to obtain $\mathbf{Z}_l$. Subsequently, $\mathbf{Z}_l$ splits into two branches: the first one passes through the up block in the encoder part, producing $\mathbf{E}_l^{\text{out}}$ for lower layers; the second one goes through the up block in the decoder part, and after fusing with the decoder features $\mathbf{D}_l^{\text{in}}$ from lower layers, it yields the output feature $\mathbf{D}_l^{\text{out}}$ for layer l.

B. Network Architecture

The network architecture is depicted in Fig. 2. It consists of L CED layers that serve as multiscale feature extractors, with each embedding a different spatial downsampling rate.

1) CED Layer: One CED layer comprises an encoder and a decoder, as described in Fig. 3. Taking layer l as an example, the input $\mathbf{E}_l^{\text{in}} \in \mathbb{R}^{H\times W\times N_{\text{bs}}}$ is fed through the cascaded

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ED structure, being spatially downsampled and restored in sequence, to output $\mathbf{D}_l^{\text{out}} \in \mathbb{R}^{H\times W\times N_{\text{bs}}}$ of this layer.

For the encoder, a spectral bottleneck consisting of two convolution blocks, followed by a residual skip, is first used to compress and restore input channels and pays more attention to distinctive spectral information. Next, there is a downsampling block that reduces the features to $\mathbf{Z}_l \in \mathbb{R}^{(\kappa_l\cdot H)\times(\kappa_l\cdot W)\times N_{\text{bs}}}$ through average pooling, shrinking the spatial dimensions by a factor of κ_l.

For the decoder, an upsampling block using bilinear interpolation and a convolution block is used to restore the spatial size. Then, a convolution block is finally appended that outputs $\mathbf{D}_l^{\text{out}}$.

For flexible interlayer connections that will be explained later, we add an extra upsampling block in the encoder that takes $\mathbf{Z}_l$ as input and outputs $\mathbf{E}_l^{\text{out}} \in \mathbb{R}^{H\times W\times N_{\text{bs}}}$. Additionally, $\mathbf{D}_l^{\text{in}} \in \mathbb{R}^{H\times W\times N_{\text{bs}}}$ from other layers is added in the decoder before the convolution block.

2) Stacking Multiscale Layers With Skip Connections: We fuse multiscale features extracted from the input image by integrating several layers with different κ values, which are respectively set to $\{1, 1/2, 1/4, \ldots, 1/2^{L-1}\}$ for L layers. A simple and natural approach is to stack the layers up and connect adjacent layers l and $l+1$ by passing $\mathbf{E}_l^{\text{out}}$ to $\mathbf{E}_{l+1}^{\text{in}}$ and $\mathbf{D}_{l+1}^{\text{out}}$ to $\mathbf{D}_l^{\text{in}}$. Then, the input and output of the multiscale network are defined as the topmost layer $\mathbf{E}_1^{\text{in}} \equiv \mathbf{X}_{\text{bs}}$ and $\hat{\mathbf{X}}_{\text{bs}} \equiv \mathbf{D}_1^{\text{out}}$, respectively. Based on this simple consideration, we introduce interlayer skip connections that make multiscale features flow more flexibly within the network, and contribute to better background reconstruction. Concretely, for each layer l, we have

$$\mathbf{E}_l^{\text{in}} = \begin{cases} \mathbf{X}_{\text{bs}}, & l = 1 \\ \mathbf{X}_{\text{bs}} + \sum_{k=1}^{f} l-1\mathbf{E}_k^{\text{out}}, & l = 2, 3, \ldots, L \end{cases} \tag{7}$$

$$\mathbf{D}_l^{\text{in}} = \begin{cases} \mathbf{0}, & l = L \\ \sum_{k=l+1}^{L} \mathbf{D}_k^{\text{out}}, & l = 1, 2, \ldots, L-1. \end{cases} \tag{8}$$

Clearly, (7) indicates that we input layer l not with the encoder output $\mathbf{E}_{l-1}^{\text{out}}$ of the upper adjacent layer, but with $\mathbf{X}_{\text{bs}}$ and the sum of the outputs $\sum_k \mathbf{E}_k^{\text{out}}$ of all previous layers. Similarly, (8) shows that the decoder input receives the sum of the outputs $\sum_k \mathbf{D}_k^{\text{out}}$ of all lower layers.

C. Loss Function

1) Multiscale Reconstruction Loss: To retain the multiscale information from deeper layers, we utilize a multiscale reconstruction loss as the content loss, which is given by

$$\mathcal{L}_{\text{MSR}} = \frac{1}{L}\sum_{l=1}^{L} \frac{1}{S\left((\mathbf{X}_{\text{bs}})_{\downarrow\kappa_l}\right)} \left\| \left(\mathbf{D}_l^{\text{out}}\right)_{\downarrow\kappa_l} - (\mathbf{X}_{\text{bs}})_{\downarrow\kappa_l} \right\|_F^2 \tag{9}$$

where $S(\cdot)$ obtains the pixel number, and $(\cdot)_{\downarrow\kappa_l}$ downsamples the image at rate κ_l over spatial dimensions. We treat all decoder outputs as downsampled images of $\mathbf{X}_{\text{bs}}$, so that those deeper outputs will retain their multiscale nature. Note that when $L = 1$, $\mathcal{L}_{\text{MSR}}$ degenerates to the mse loss.

2) Enhanced Separation Training Loss: The separation training strategy can effectively suppress anomalies through the utilization of a binary mask mechanism, but it, however, relies heavily on the availability of a precise threshold value. On the one hand, the proportion of anomalous pixels in one image is extremely imbalanced, which causes accurate prediction to be difficult. On the other hand, the distribution of background and anomalies varies significantly across different datasets, suggesting that we can hardly develop a unified threshold prediction method that generalizes well to unseen datasets. For this reason, here we propose an enhanced version of the separation training algorithm in order to completely remove the impact of any threshold-related parameter.

Our strategy is to "soften" the hard thresholding while preserving the characteristics of separating anomalies from background during training. First, we calculate $\mathbf{M}$ by min–max scaling to replace (6)

$$\mathbf{M}_{i,j} = \frac{\mathbf{R}_{i,j} - \text{Min}(\mathbf{R})}{\text{Max}(\mathbf{R}) - \text{Min}(\mathbf{R})} \tag{10}$$

where $\text{Min}(\cdot)$ and $\text{Max}(\cdot)$ obtain the minimum and maximum values of the matrix, respectively. Equation (10) indicates that we allow $\mathbf{M}$ to be a soft separator, where the separation between background and anomaly is less strict compared to the previous hard one, but no longer needs binarization. Accordingly, the complementary term $\bar{\mathbf{M}}$ of $\mathbf{M}$ is extended as $\bar{\mathbf{M}} = \mathbf{1} - \mathbf{M}$ instead of Boolean operations. We denote the softened separation training loss as $\mathcal{L}_{\text{SeT+}}$, which is calculated as follows:

$$\mathcal{L}_{\text{SeT+}} = \mathcal{L}_{\text{BR}} + \lambda \cdot \mathcal{L}_{\text{AS}} \tag{11}$$

where $\mathcal{L}_{\text{BR}}$ and $\lambda \cdot \mathcal{L}_{\text{AS}}$ are the background reconstruction term and anomaly suppression term introduced in (1) and (2), respectively. λ is a tradeoff parameter. The calculation of $\mathcal{L}_{\text{SeT+}}$ here is similar as $\mathcal{L}_{\text{SeT}}$ in (3). The difference is that $\mathbf{M}$ and $\bar{\mathbf{M}}$ calculated by (10) here are real matrices, rather than Boolean matrices.

3) Total Loss: The total loss function consists of the above two parts, given as the following:

$$\mathcal{L}_{\text{total}} = \frac{1}{N_{\text{bs}}}(\mathcal{L}_{\text{MSR}} + \mathcal{L}_{\text{SeT+}}). \tag{12}$$

D. Training and Detection

The enhanced separation training algorithm is summarized in Algorithm 2. We set K and E to 5 and 150, respectively. ADAM is chosen as the optimizer, and the learning rate is set to 0.001. After training, we can directly obtain $\hat{\mathbf{X}}_{\text{bs}} = \mathbf{D}_1^{\text{out}}$ by network inference. The detection result $\mathbf{R}$ is computed by the following:

$$\mathbf{R}_{i,j} = \|(\hat{\mathbf{X}}_{\text{bs}})_{i,j,:} - (\mathbf{X}_{\text{bs}})_{i,j,:}\|_2^2 \tag{13}$$

where $(\cdot)_{i,j,:}$ is the spectral vector at position (i, j).

IV. Experiments

In this section, we first give a brief introduction to the basic experimental settings we followed in Section IV-A.

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Algorithm 2 Enhanced Separation Training Algorithm

Input: HSI data $\mathbf{X}_{bs}$ after band selection, tradeoff hyperparameter $\lambda > 0$, number of training iterations $K \geq 1$, number of epochs in one iteration $E \geq 1$.

Initialize: $\mathbf{M}^{(0)} = \mathbf{0}$.

```
for m = 1 : K do
    for n = 1 : E do
        Feed the network with X_bs and yield X̂_bs^(m);
        Calculate the total loss using (12);
        Do backward propagation;
    end for
    Calculate R_{i,j}^(m) = ‖(X̂_bs^(m))_{i,j,:} − (X_bs^(m))_{i,j,:}‖_2^2;
    Calculate M^(m) using (10);
end for
```

Output: Detection map $\mathbf{R}^{(K)}$.

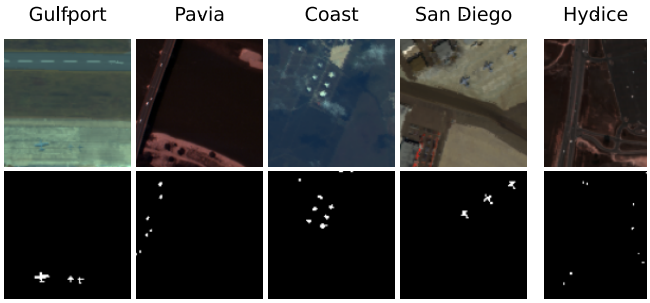


Fig. 4. (Upper row) Pseudocolor images of the classic datasets and (lower row) their corresponding ground truth maps.

Then, comprehensive comparisons with competitors are presented in Section IV-B. To verify the individual effectiveness of the network structure and training process in our proposed self-supervised method, we discuss them separately in Sections IV-C and IV-D. Finally, the analysis of important hyperparameters is given in Section IV-E.

A. Experimental Settings

1) Implementation: Our method was implemented using PyTorch code, and run on a PC equipped with an Intel Core i7-11800H CPU at 2.30 GHz with 16 GB RAM, and an NVIDIA 3070 GPU with 8 GB of memory.

2) Datasets: We used six datasets including five classic datasets Gulfport, Pavia, Coast, San Diego, and Hydice, as well as the larger one HAD100 for evaluation. The pseudocolor images and ground truth maps of the classic datasets are shown in Fig. 4 and part of the HAD100 dataset is shown in Fig. 5. More detailed information is listed as follows.

1) Gulfport was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. It has a size of $100 \times 100 \times 191$.
2) Pavia was collected using the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. It has a size of $150 \times 150 \times 102$.
3) Coast was also acquired from the AVIRIS sensor. It contains 100×100 pixels and 204 spectral bands.
4) San Diego is a subset of size $100 \times 100 \times 189$ selected from an entire image obtained by the AVIRIS sensor.

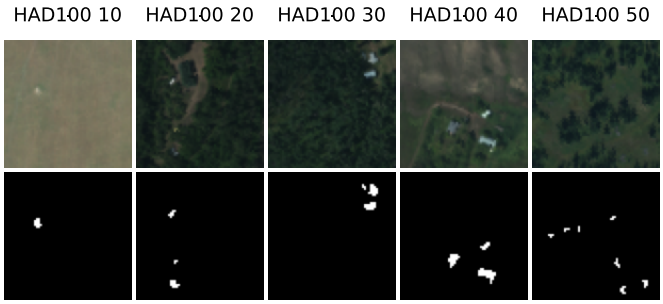


Fig. 5. (Upper row) Pseudocolor images of the HAD100 dataset and (lower row) their corresponding ground truth maps. We only display scenes 10, 20, 30, 40, and 50 from the HAD100 dataset for visualization here.

5) Hydice was collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. It has a spatial size of 100×80 and 162 spectral bands.
6) HAD100 [44] was collected by the AVIRIS sensor. It contains 100 scenes of size $64 \times 64 \times 276$ after removing bad bands from the original 425.

3) Metrics: We used the receiver operating characteristic (ROC) curves of parameter pairs (P_d, P_f) and (P_f, τ), along with the corresponding area under the curve (AUC) values, to evaluate a detection algorithm. The former (P_d, P_f) curve contains detection probabilities P_d on all false alarm rates P_f tested from 0 to 1. A model with stronger anomaly-background separability exhibits a curve that approaches the upper left corner, consequently yielding a higher AUC(P_d, P_f) close to 1. Besides, an AUC(P_f, τ) closer to 0 suggests low false alarm rates over a range of threshold values τ, signifying fewer background pixels incorrectly classified as anomalous samples and thus reflecting a higher background reconstruction capability. In addition, we recorded the run times as another quantitative metric. For a more intuitive presentation, we visualized the detection maps of an included method.

B. Performance Comparisons

1) Comparative Methods: We chose seven methods, including RXD [15], CRD [22], LRASR [23], LREN [40], RGAE [41], Auto-AD [42], and BiGSeT [46] as benchmarks for comparison. RXD, CRD, and LRASR are the most representative model-driven methods, while the remaining ones are data-driven detectors. The RXD is a probability distribution-based method; the CRD and LRASR are both representation-based methods; LREN detects the anomalies using encoded hidden features; and RGAE, Auto-AD, and BiGSeT reconstruct the background in a self-supervised manner. For our method, we set the regularization parameter λ to 10^{-4}, 10^{-3}, 10^{-3}, 1 and 10^{-4}, respectively, for the Gulfport, Pavia, Coast, San Diego, and Hydice datasets, and fixed it to 10^{-3} for all images in the HAD100. The band number N_{bs} and layer number L were fixed at 64 and 3 for all datasets.

2) Quantitative Comparisons: Three quantitative metrics for different methods on the classic datasets are listed in Table I. The best and second best results are marked in **bold** and underlined, respectively. It can be seen that our method achieved the highest scores of AUC(P_d, P_f) on these datasets,

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TABLE I

AUC OF (P_d, P_f), AUC OF (P_f, τ), AND RUN TIMES (IN SECONDS) FOR DIFFERENT METHODS ON CLASSIC DATASETS

Metric	Dataset	Ours[1]	RXD 1990 [15]	CRD 2015 [22]	LRASR 2016 [23]	LREN 2021 [40]	RGAE 2022 [41]	Auto-AD[1] 2022 [42]	BiGSeT[1] 2023 [46]
AUC(P_d, P_f) ↑	**Gulfport**	0.9954	0.9526	0.9438	0.9202	0.7375	0.7531	0.9690	**0.9966**
	Pavia	**0.9976**	0.9538	0.9722	0.9543	0.8673	0.9042	0.9900	0.9871
	Coast	**0.9980**	0.9907	0.9961	0.9394	0.9489	0.9822	0.9882	0.9915
	San Diego	**0.9933**	0.8885	0.9683	0.9681	0.9487	0.9899	0.8445	0.9635
	Hydice	**0.9999**	0.9763	0.9911	0.9939	0.9698	0.8366	0.9948	0.9976
AUC(P_f, τ) ↓	**Gulfport**	**0.0014**	0.0248	0.0202	0.0598	0.2545	0.0754	0.0271	0.0062
	Pavia	**0.0004**	0.0199	0.0380	0.0310	0.0926	0.0444	0.0021	0.0131
	Coast	**0.0016**	0.0556	0.0161	0.1141	0.1630	0.0168	0.0071	0.0020
	San Diego	0.0602	0.0383	0.0435	0.0258	0.1972	0.0174	0.0120	**0.0033**
	Hydice	**0.0041**	0.0372	0.0197	0.0958	0.1849	0.0909	0.0081	0.0070
Time (s) ↓	**Gulfport**	9.82	**0.10**	23.72	31.80	87.00	58.42	3.73	10.91
	Pavia	17.56	**0.15**	52.98	74.51	167.25	92.02	27.23	13.37
	Coast	9.66	**0.11**	22.62	34.18	89.74	61.91	6.44	11.64
	San Diego	9.83	**0.11**	23.83	27.37	86.39	57.95	20.00	12.64
	Hydice	6.96	**0.08**	21.01	24.15	65.37	21.20	20.36	3.40

[1] These methods were run on the GPU, while others were implemented on the CPU.

TABLE II

AUC OF (P_d, P_f), AUC OF (P_f, τ), AND RUN TIMES (IN SECONDS) FOR DIFFERENT METHODS ON HAD100 DATASET

Metric	Ours[1]	RXD 1990 [15]	CRD 2015 [22]	LRASR 2016 [23]	LREN 2021 [40]	RGAE 2022 [41]	Auto-AD[1] 2022 [42]	BiGSeT[1] 2023 [46]
AUC(P_d, P_f) ↑	**0.9819** ± 0.0283	0.9583 ± 0.0555	0.9177 ± 0.1017	0.9522 ± 0.0631	0.8838 ± 0.1317	0.8366 ± 0.1349	0.8951 ± 0.0894	0.9742 ± 0.0385
AUC(P_f, τ) ↓	**0.0037** ± 0.0079	0.0509 ± 0.0352	0.0170 ± 0.0094	0.0560 ± 0.0420	0.1753 ± 0.0769	0.0572 ± 0.0452	0.0132 ± 0.0162	0.0046 ± 0.0054
Time (s) ↓	8.45	**0.03**	1.09	10.67	32.20	11.47	12.11	1.37

[1] These methods were run on the GPU, while others were run on the CPU.

which indicates superior probabilities of detection. The scores of AUC(P_f, τ) for our method on Gulfport, Pavia, Coast, and Hydice datasets are very close to 0, suggesting minimal false alarm rates across a broad range of threshold values. Additionally, the run times of our method are relatively low and acceptable, compared to other representation-based and deep learning-based methods.

We conducted a similar experiment on the HAD100 dataset. Before testing on each image, we applied the OPBS to reduce the original channel number to 64 because it is too time-consuming for most methods. In this setting, each input image has a size of $64 \times 64 \times 64$. We computed the mean ($\pm$std) values of AUC(P_d, P_f) and AUC(P_f, τ) of 100 images. The results are listed in Table II. It can be observed that our method obtained the best values of AUC(P_d, P_f) and AUC(P_f, τ) among others statistically. The average run time for ours is lower than the convolutional structure Auto-AD under the same settings.

To conclude, the performance comparisons on six datasets demonstrate the superiority of our method over its counterparts in terms of detection performance under different scenes with different anomalies, and architecture complexity.

3) Visual Comparisons: We display the detection maps for different methods on the classic datasets in Fig. 6. It can be observed that our method highlighted the desired objects in different scenes clearly and accurately, while effectively suppressing the background. However, other methods failed to do so. For example, the RXD, CRD, and RGAE failed to detect all airplanes on Gulfport dataset, while the LRASR, LREN, and Auto-AD wrongly focused on most background components. The ROC curves of (P_d, P_f) and (P_f, τ) for different methods on Pavia, Coast, and Hydice datasets are, respectively, shown in Figs. 7 and 8. The curves of (P_d, P_f) for our method (in black) cover all other methods, suggesting higher AUC(P_d, P_f) scores. Meanwhile, our false alarm rates are quite close to 0 across a wide range of threshold values as shown in Fig. 8, indicating very low AUC(P_f, τ) scores.

We present the separation performance between background and anomalous targets in Fig. 9. The normalized separability maps obtained by all methods on San Diego and Hydice datasets are illustrated. A separability map reflects the distribution of the background and anomalies in a detection map. The vertical dashed line spans the maximum and minimum ranges of the numerical distribution. The box contains a data

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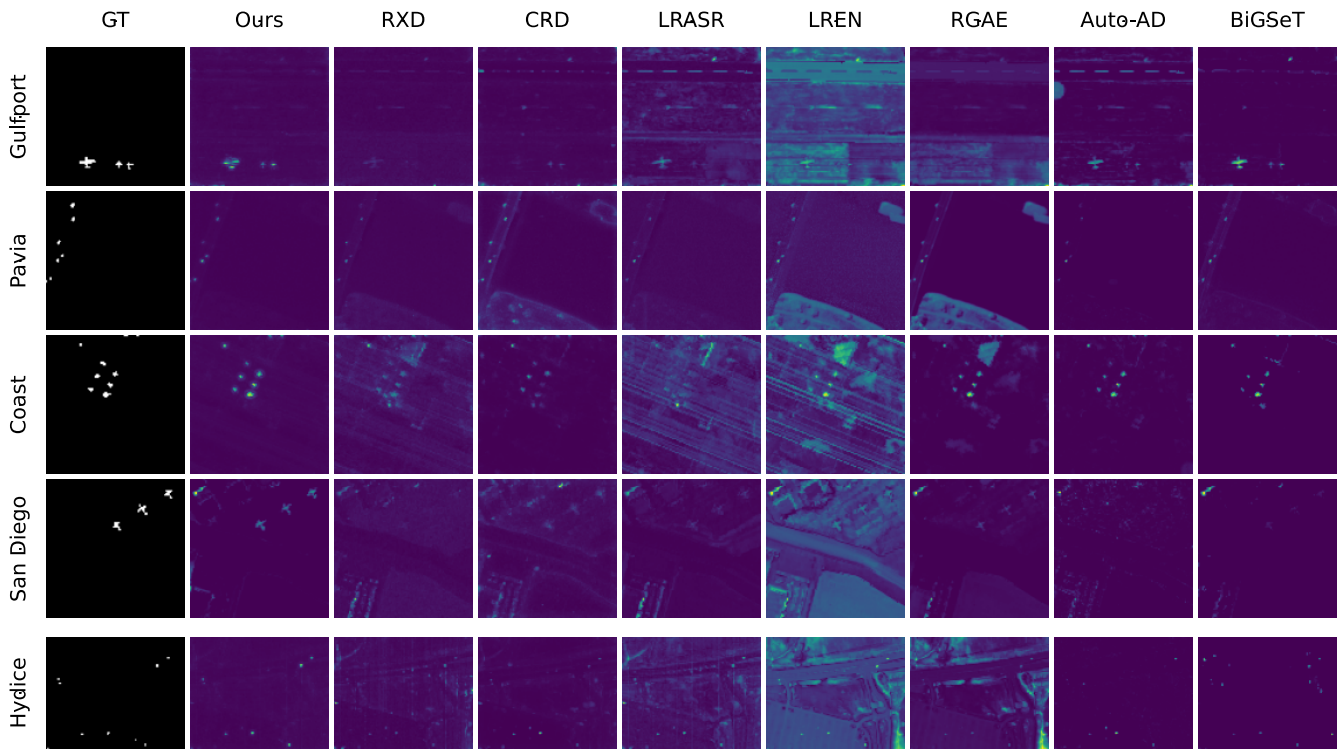


Fig. 6. (First column) Ground truth maps of five datasets and (remaining columns) corresponding detection maps for different methods.

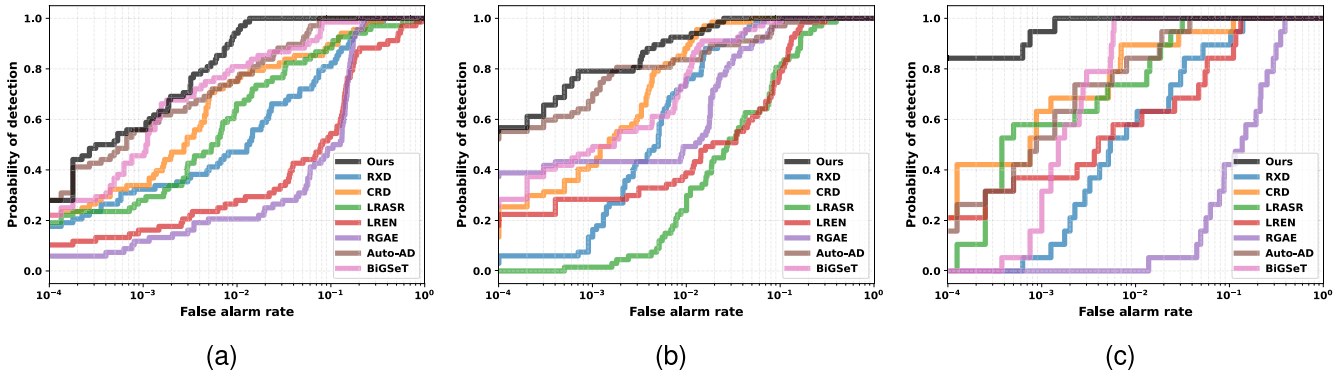


Fig. 7. ROC curves of (P_d, P_f) for different methods on (a) Pavia, (b) Coast, and (c) Hydice datasets.

range from 10% to 90%. The horizontal black line in the box represents the median of the data. The red and green colors denote the anomaly and background, respectively. Obviously, a method has a superior separation performance when its anomaly box leaves a large gap from the background box at the bottom. From Fig. 9, we can observe that the anomalies are separated from at least 90% of the background on the San Diego and Hydice datasets, while other methods cannot. Note that our maximum background value on the Hydice dataset is the smallest among all methods, indicating the best background anomaly separability.

C. Validation of Network Structure

We first validate the effectiveness of our proposed network structure MSNet on the classic datasets, by comparing it with two convolutional HAD frameworks Auto-AD [42] and MSBRNet [43]. The former is an encoder–decoder structured network, and the latter is composed of a low-rank module and a multiscale module. We set $L = 3$ and that means MSNet comprises three CED layers in this experiment. For a fair comparison, we maintained all other variables constant, except for the network structure. For example, we did not include band selection in our method prior to training. And these three networks were all trained using $\mathcal{L}_{\text{SeT}+}$ with $\lambda = 10^{-2}$. The AUC scores and run times are recorded in Table III. We can conclude from the table that our network can obtain the highest AUC scores in most cases while requiring the lowest run times at the same time.

Furthermore, we conducted a comparative experiment on the HAD100 dataset. The well-known multiscale network architecture U-Net[2] [47] was used in our discussion here.

[2]A public PyTorch implementation is available at https://github.com/milesial/Pytorch-UNet.

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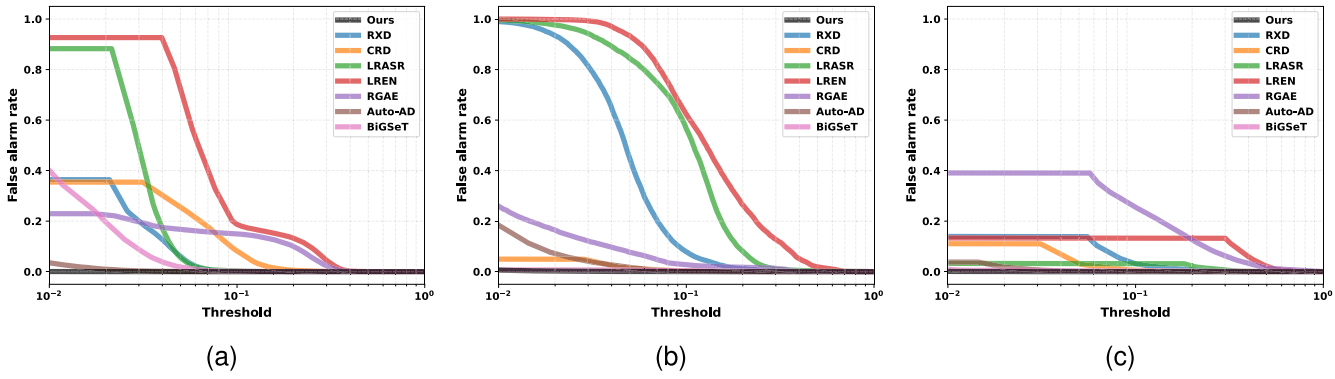


Fig. 8. ROC curves of (P_f, τ) for different methods on (a) Pavia, (b) Coast, and (c) Hydice datasets.

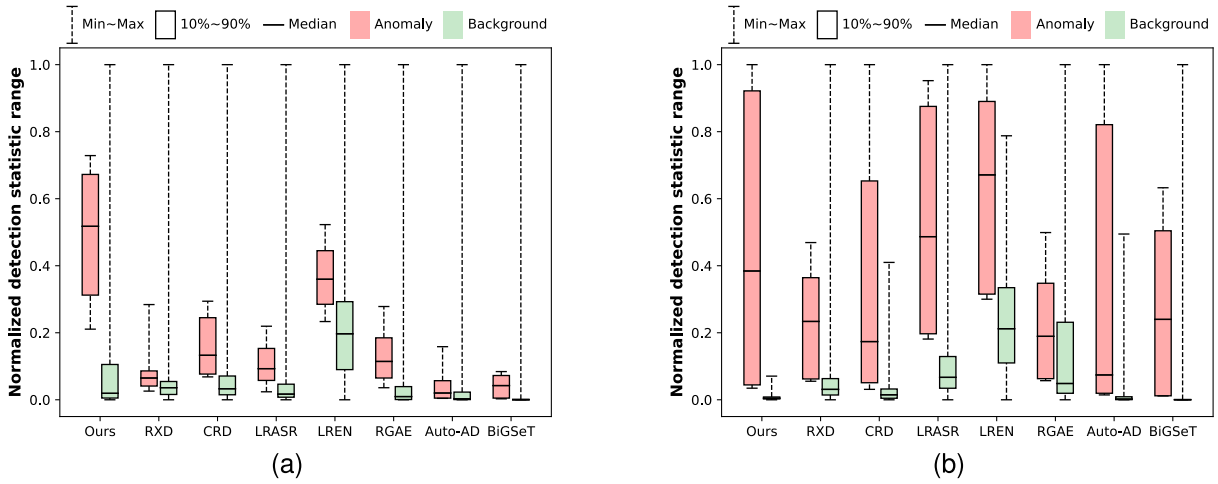


Fig. 9. Background-anomaly separability results for different methods on (a) San Diego and (b) Hydice datasets.

TABLE III

COMPARISON OF NETWORKS MSNET, AUTO-AD, AND MSBRNET ON CLASSIC DATASETS USING AUC OF (P_d, P_f) AND RUN TIMES (IN SECONDS) AS METRICS

Metric	Network	Gulfport	Pavia	Coast	San Diego	Hydice
AUC(P_d, P_f) ↑	**MSNet**	**0.9945**	**0.9904**	**0.9917**	0.9855	**0.9883**
	Auto-AD	0.9924	0.9887	0.9916	**0.9889**	0.9837
	MSBRNet	0.9897	0.9802	0.9713	0.8381	0.9715
Time (s) ↓	**MSNet**[1]	**23.15**	**26.18**	**25.11**	**22.73**	**10.77**
	Auto-AD[1]	135.48	149.69	165.36	144.92	124.53
	MSBRNet[1]	38.66	44.28	43.34	43.86	36.74

[1] All three networks were run on the GPU.

TABLE IV

COMPARISON OF NETWORKS MSNET AND U-NET ON HAD100 DATASET

Metric	MSNet[1]	U-Net[1]
AUC(P_d, P_f) ↑	**0.9819** ± 0.0283	0.9779 ± 0.0323
Time (s) ↓	**8.45**	15.63

[1] All these methods were run on the GPU.

We compared the performance of MSNet and U-Net with the $\mathcal{L}_{\text{SeT}+}$ under the same settings introduced in Section IV-B to validate the effectiveness of our network. Comparing MSNet with U-Net, as shown in Table IV, we observe that our MSNet achieved a better AUC(P_d, P_f) value, indicating that MSNet architecture is statistically more effective for HAD tasks than U-Net. On the other hand, the computational complexity of MSNet is smaller than U-Net. This is because MSNet retains the input channels throughout all downsampling branches. In contrast, U-Net increases channel numbers to preserve more channel-related information, resulting in a large number of parameters that are unnecessary for HAD tasks.

In conclusion, our multiscale network can perform better than other architectures across six datasets that contain different spectral sensors and anomalous targets, confirming its adaptability and robustness while maintaining low computational complexity.

D. Validation of Enhanced Separation Training

To focus on discussing the proposed enhanced version of separation training, we set a vanilla AE structure as the detection network for both training Algorithms 1 (denoted as $\mathcal{L}_{\text{SeT}}$ for simplicity) and 2 (denoted as $\mathcal{L}_{\text{SeT}+}$), and compared their detection performance. The AE comprises a linear layer

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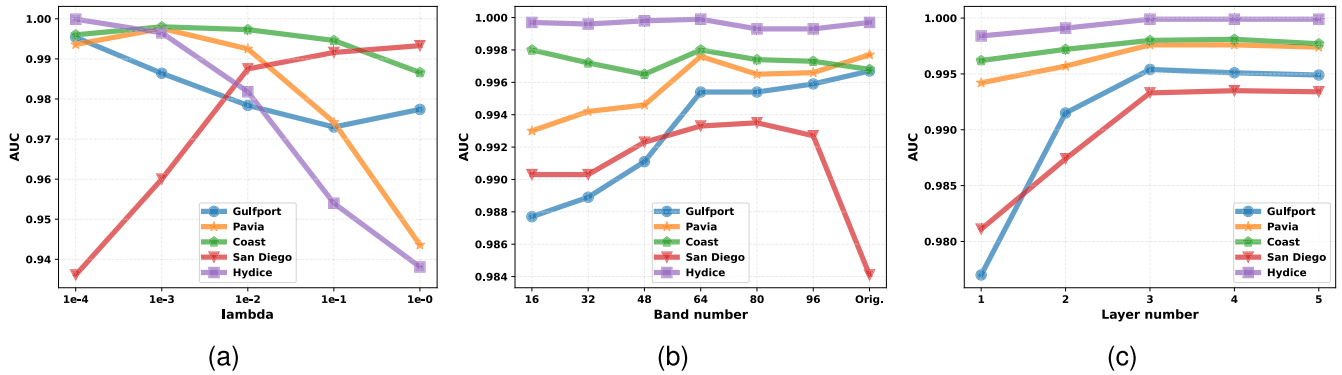


Fig. 10. Analysis of hyperparameters using AUC of (P_d, P_f) as metric on the classic datasets. (a) Regularization parameter λ. (b) Band number N_{bs}. (c) Layer number L. Datasets Gulfport, Pavia, Coast, San Diego, and Hydice are, respectively, shown as blue, orange, green, red, and purple curves.

TABLE V

COMPARISON OF ALGORITHMS $\mathcal{L}_{\text{SeT}}$ AND $\mathcal{L}_{\text{SeT+}}$ WHEN APPLIED TO THE VANILLA AE ON CLASSIC DATASETS USING AUC OF (P_d, P_f) AS METRIC

Algorithm Ver.	Gulfport	Pavia	Coast	San Diego	Hydice
$\mathcal{L}_{SeT}$	**0.9966**	0.9871	0.9915	0.9635	0.9976
$\mathcal{L}_{SeT+}$	0.9964	**0.9876**	**0.9943**	**0.9704**	**0.9981**

TABLE VI

COMPARISON BETWEEN WITH AND WITHOUT THE ENHANCED SEPARATION TRAINING WHEN APPLIED TO MSNET AND U-NET ON HAD100 DATASET USING AUC OF (P_d, P_f) AS METRIC

Network	w/ $\mathcal{L}_{SeT+}$	w/o $\mathcal{L}_{SeT+}$
MSNet	**0.9819** $\pm$ 0.0283	0.9546 $\pm$ 0.0510
U-Net	**0.9779** $\pm$ 0.0323	0.8904 $\pm$ 0.1325

with 100 hidden units, a ReLU activation function, and a linear layer that restores the input channels. The detection results tested on five datasets including Gulfport, Pavia, Coast, San Diego, and Hydice are listed in Table V. It can be seen that the $\mathcal{L}_{\text{SeT+}}$ performs better than $\mathcal{L}_{\text{SeT}}$ on average, indicating that the enhanced version $\mathcal{L}_{\text{SeT+}}$ can efficiently suppress the anomalies while training.

Additionally, we conducted experiments on the HAD100 dataset, comparing the results with and without the enhanced separation training strategy when the detection networks are set as MSNet and U-Net. The results are listed in Table VI. We can find that enabling $\mathcal{L}_{\text{SeT+}}$ leads to significant performance improvements for both networks, providing clear evidence for the effectiveness of the enhanced separation training. Moreover, the training algorithm is convenient to use because we do not need to finely adjust the threshold hyperparameter for 100 images, which however is critical for the BiGSeT.

In conclusion, our enhanced separation training algorithm exhibits its effectiveness and advantage over the original version on six datasets, in terms of adaptability and automation.

E. Hyperparameter Analysis

In this section, we first analyze how hyperparameters including λ, N_{bs}, and L impact the detection performance of our model. The scores of AUC(P_d, P_f) regarding these three factors are shown in Fig. 10(a)–(c). We then conduct an ablation study for other main components.

1) Regularization Parameter λ: λ controls the effect of anomaly suppression. We set $N_{\text{bs}} = 64$ and $L = 3$ and chose λ values from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. As shown in Fig. 10(a), the performance of five datasets starts to decrease after reaching the maximum, except for the San Diego dataset (in red) which keeps on increasing. The optimal values for the five datasets are 10^{-4}, 10^{-3}, 10^{-3}, 1, and 10^{-4}, respectively.

2) Band Number N_{bs}: N_{bs} is the number of band subset selected by the OPBS [48], which also indicates the input channels of the network. We set λ to the optimal and fix $L = 3$, and compared N_{bs} values chosen from $\{16n\}_{n=1}^{6}$, as well as the original band number without band selection (denoted as Orig., see Section IV-A). The AUC results are displayed in Fig. 10(b). It is evident that, across the five tested datasets, a subset of the original bands (bands 64–96) can be adequate to achieve high performance.

Moreover, it can be seen from Table VII that the run times increase as N_{bs} gets larger from 16 to 96. However, it is noteworthy that the run times for all these cases are still lower than Orig. This is because OPBS can save time for the deep network by reducing the number of input channels, and the process of performing OPBS itself does not require much time. In conclusion, band selection can help maintain (or improve) the AUC performance while reducing the time cost for the network.

We suggest a proper value of 64 for N_{bs} considering the AUC performance and time cost.

3) Layer Number L: L is the number of multiscale CED layers that we stack. We set λ to the optimal and fix $N_{\text{bs}} = 64$, and studied the effects of L chosen from 1 to 5. The trends of AUC are shown in Fig. 10(c) and the run times are tabulated in Table VIII. It can be observed that the AUC scores tend to increase as L gets larger. This is because more multiscale features lead to better background estimation and thus improve detection performance.

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TABLE VII
RUN TIMES (IN SECONDS) OF BAND SELECTION/TOTAL COST FOR DIFFERENT BAND NUMBERS ON THE CLASSIC DATASETS

Band No.	Gulfport	Pavia	Coast	San Diego	Hydice
16	0.28 / 6.35[1]	0.41 / 6.98	0.31 / 6.42	0.27 / 6.38	0.15 / 5.97
32	0.51 / 7.07	0.75 / 10.38	0.56 / 7.07	0.50 / 7.08	0.29 / 6.37
48	0.72 / 8.76	1.02 / 14.01	0.82 / 7.98	0.73 / 8.31	0.41 / 6.82
64	0.93 / 9.82	1.29 / 17.56	1.06 / 9.66	0.94 / 9.83	0.51 / 6.96
80	1.16 / 11.56	1.39 / 21.76	1.25 / 11.42	1.14 / 11.54	0.60 / 7.18
96	1.26 / 13.35	1.46 / 25.44	1.43 / 13.27	1.30 / 13.41	0.67 / 7.51
Orig.	0.00 / 23.15	0.00 / 26.18	0.00 / 25.11	0.00 / 22.73	0.00 / 10.77

[1] Total time cost includes band selection, training and inference.

TABLE VIII
RUN TIMES (IN SECONDS) FOR DIFFERENT LAYER NUMBERS ON THE CLASSIC DATASETS

Layer No.	Gulfport	Pavia	Coast	San Diego	Hydice
1	6.43	12.28	6.24	6.46	3.14
2	8.09	14.86	7.89	8.14	5.05
3	9.82	17.56	9.66	9.83	6.96
4	11.71	20.35	11.44	11.73	9.17
5	13.81	24.07	13.78	14.04	10.90

TABLE IX
ABLATION STUDY FOR OTHER MAIN COMPONENTS USING AUC OF (P_d, P_f) AS METRIC

$\mathcal{L}_{SeT+}$[1]	$\mathcal{L}_{MSR}$[1]	$\mathcal{SC}$	Gulfport	Pavia	Coast	San Diego	Hydice
✗	✗	✗	0.9794	0.9875	0.9841	0.9538	0.9842
✓	✗	✗	0.9906	0.9951	0.9974	0.9929	0.9983
✓	✓	✗	0.9934	0.9966	0.9977	0.9930	0.9986
✓	✓	✓	**0.9954**	**0.9976**	**0.9980**	**0.9933**	**0.9999**

[1] When neither $\mathcal{L}_{SeT+}$ nor $\mathcal{L}_{MSR}$ is used, we train the network with only the MSE loss.

However, adding layers results in increases in computational cost, as shown in Table VIII, but does not lead to an unlimited improvement in detection performance. Notably, we do not need more layers than three, because the AUC performance starts to get saturated when $L \geq 3$.

Therefore, we suggest an appropriate value of 3 for L as a good tradeoff between the AUC performance and time cost.

4) Other Main Components: The considered components are two loss functions $\mathcal{L}_{\text{SeT+}}$ and $\mathcal{L}_{\text{MSR}}$, as well as the skip connections (denoted as $\mathcal{SC}$). We studied their effects by incorporating them into the model sequentially. The results are listed in Table IX. We can conclude that each one contributed to an improved performance, thus validating their respective effectiveness.

V. CONCLUSION

In this article, we propose a multiscale network architecture MSNet with enhanced separation training for self-supervised HAD. Considering the anomalies may exhibit different scales, the network is built with several multiscale CED layers and stacked up using skip connections to adapt to different situations. The network is trained by the multiscale reconstruction loss that retains the multiscale features, and the enhanced separation training loss that suppresses potentially included anomalies in the estimated background in a more automated manner. We detect anomalies by performing a network inference and computing the reconstruction errors after training. Moreover, before training and detection, a fast and efficient band selection algorithm OPBS is adopted to reduce redundant bands and decrease the computational complexity. Extensive experiments conducted on six datasets show that our method surpasses some comparative methods in terms of overall detection performance and computational complexity. Hyperparameter analysis validates the effectiveness of the main components and provides a better understanding of them.

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